PAPER 1

1. Introduction

This project focuses on the classification of Alzheimer's disease stages using deep learning. The model was trained to identify different levels of cognitive impairment based on brain scan images.

2. Research Paper Reference

We referred to multiple research papers on Alzheimer's disease classification using CNNs. The primary reference was:

- "Deep Learning for Alzheimer's Disease Detection" - This paper helped in designing our convolutional neural network (CNN) architecture and selecting preprocessing techniques.

3. Data Preprocessing & Training

The dataset consisted of brain scan images categorized into four classes:

- Mild Impairment
- Moderate Impairment
- No Impairment
- Very Mild Impairment

We applied the following preprocessing steps:

- Resizing images to 224x224 pixels
- Normalizing pixel values
- Augmenting data to improve generalization

The model was trained locally on a PC using the PyTorch framework. The training process involved multiple epochs, with the Adam optimizer and cross-entropy loss function.

4. Model Evaluation

After training, the model was evaluated using a validation dataset. The evaluation metrics included:

- **Validation Loss**: 0.6099
- **Accuracy**: 71.46%

5. GUI Implementation

A PyQt5-based GUI was developed, allowing users to:

- Upload an image for classification
- View the model's prediction in real-time

6. Conclusion

The project successfully trained and evaluated an Alzheimer's disease classification model locally on a PC. The integration of a user-friendly GUI enables easy image classification. Future improvements may include model fine-tuning and dataset expansion to improve accuracy.

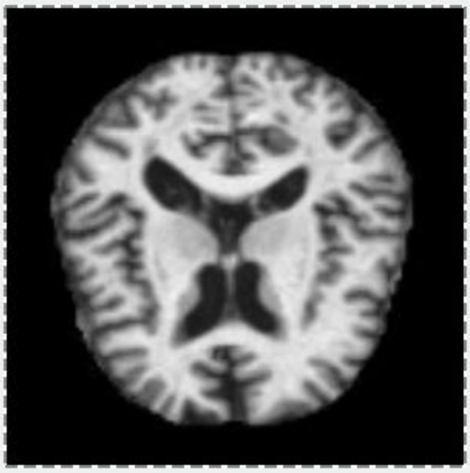
Final Status: Model training and evaluation completed locally. GUI fully integrated for real-time image classification.

```
PS E:\Intern\MEGA-MINDS> & "C:/Users/Himanshu Gaur/AppData/Local/Programs
Using device: cpu
Epoch 1/10, Loss: 0.9510
Epoch 2/10, Loss: 0.6087
Epoch 3/10, Loss: 0.5411
Epoch 4/10, Loss: 0.5022
Epoch 5/10, Loss: 0.4628
Epoch 6/10, Loss: 0.4332
Epoch 7/10, Loss: 0.4159
Epoch 8/10, Loss: 0.3974
Epoch 9/10, Loss: 0.3756
Epoch 10/10, Loss: 0.3567
Training complete!
Model saved as E:\Intern\MEGA-MINDS\Trained Model\alzheimers_model.pth
```

```
PS E:\Intern\MEGA-MINDS> & "C:/Users/Himanshu Gaur/Ap
Using device: cpu

✅ Model already trained. Loading saved model...
Using device: cpu
Using device: cpu
Validation Loss: 0.6099, Accuracy: 71.46%
Model evaluation complete!
PS E:\Intern\MEGA-MINDS> 
```

Alzheimer's Disease Classifier



Upload Image

Prediction: Moderate Impairment